Guided practice: observation and judgement

Circle one answer in each set of brackets; use boxes for multiple choices. Each written answer has two lines.

Initial oil + water



B1.1 Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?

Describe what you can see. Keep explanations such as “digested” separate.

B1.2 Evaluate this given prediction and reason. You do not need to invent an “original hypothesis”.

“With the other conditions kept the same, adding 2 mL Y will spread large oil droplets into small droplets, because all digestive enzymes emulsify lipids (乳化脂質).”

Judge separately whether observations can test the prediction and whether the reason fits the concepts. Then explain the problem with the reason.

Can observations test it? （Yes ／ No）

Does its reason fit the concepts? （Yes ／ No）

Brief reason / explanation:

B1.3 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Use the method in A. More than one factor may belong to a group.

Independent variable (the factor changed on purpose)	Dependent variable (the result you measure or observe)	Controlled variables (the factors kept the same)
☐ Total volume of oil and water	☐ Total volume of oil and water	☐ Total volume of oil and water
☐ Reaction temperature	☐ Reaction temperature	☐ Reaction temperature
☐ Reaction time	☐ Reaction time	☐ Reaction time
☐ Total volume of liquid added	☐ Total volume of liquid added	☐ Total volume of liquid added
☐ Method of shaking	☐ Method of shaking	☐ Method of shaking
☐ Appearance of the mixture	☐ Appearance of the mixture	☐ Appearance of the mixture
☐ Combination of digestive juices	☐ Combination of digestive juices	☐ Combination of digestive juices

B1.4 Choose all the conditions needed for a fair comparison.

An assumption is a condition taken as true when explaining results. The procedure must also follow the fixed conditions provided.

- ☐ Observe all setups after the same reaction time.
- ☐ The X + Y group can have more liquid in total to compare it with the other groups.
- ☐ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.
- ☐ Use the same method of shaking for every setup.

Guided practice: controls, labels and fairness

Common comparison conditions

Each set starts with the same amounts of oil and water, receives 2 mL in total, reacts at 37°C, and uses equal observation time and shaking. Recipes: control, 2 mL water; X, 2 mL X; Y, 2 mL Y; X + Y, 1 mL each. Use the researcher’ s common starting amounts, timing and shaking details; do not invent values missing from the model.

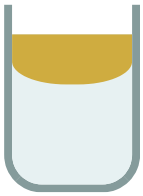
B2.1 To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?

Choose the control for this specific question. A change in another condition must not be mistaken for the effect of the juice.

B2.2 The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.

Each setup receives 2 mL in total and starts with the same amounts of oil and water.

Setup 1



Setup 2



Setup 3



Setup 4



B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

First find the total volume added to each set. Then check whether only one factor changed.

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: four main sets

B3.1 Independent application: observe the four static images and record their appearances.

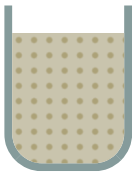
You may keep using A. Do not request the answer key yet. The images are model data, not direct product tests.

These images are static data from the current teaching model. Appearance answers are not filled in. They are not direct product-test results.

Control group: add 2 mL of water



Add 2 mL of digestive juice X



Add 2 mL of digestive juice Y



Add 1 mL each of digestive juices X and Y



Setup / conditions	Your appearance record (one per set)	
Control group: add 2 mL of water	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice X	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 1 mL each of digestive juices X and Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear

B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).

Digestive juice X mainly causes lipids to form （a clear solution ／ a cloudy emulsion (乳状液)）.

Brief reason / explanation:

B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to （decrease ／ increase）, which helps digestive juice Y act.

Brief reason / explanation:

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: heating and evidence

Boil X or Y only. Cool to 37°C, keeping starting and total added volumes, time and shaking equal.

B4.1 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

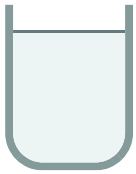
We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?（Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed. ／ A clear mixture completely proves that lipids have broken down into glycerol (甘油) and fatty acids (脂肪酸).）

Brief reason / explanation:

B4.2 To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?

B4.3 Record the appearances of the two boiled-and-cooled sets and compare them with unboiled X + Y.

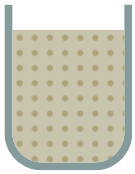
Add 1 mL each of digestive juices X and Y



Add 1 mL each of digestive juices X and Y (boil X, then cool)



Add 1 mL each of digestive juices X and Y (boil Y, then cool)



Setup / conditions	Your appearance record (one per set)	
Add 1 mL each of digestive juices X and Y (boil X, then cool)	☐ become cloudy	☐ become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	☐ become cloudy	☐ become clear

B4.4 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Extension conclusion: the digestive juice that loses its function after boiling is（Digestive juice X ／ Digestive juice Y）.

Brief reason / explanation:

B4.5 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.